Problem set 4) Conservation of energy

Data

Car m = 1270kg

V_i = 88.5 km/h = 55 mph

$88.5\ km/h = \frac{1000\ m}{1\ km} \cdot \frac{1\ hour}{3600\ s} = \frac{88.5 \cdot 1000\ m}{3600\ s} = 24.58\ m/s$

Final height?

$(1270kg)(9.8) = \frac{1}{2}(1270)(24.58)^2$

$12,446 = \frac{767304.028}{2}$

$\frac{12,446}{12,446} = \frac{383652.0}{12,446}$

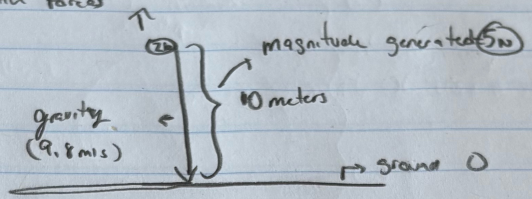
h = 30.82 meters

Problem 5 → Non Conservative Forces

Data

m = 2kg

h = 10 m



a.) = PE = ME

PE = mgh = (2)(10)(9.8)

PE = 196 J

B.) −5 = Net oppose

W = F · h =

W = −5 · 10 = −50J

(C.)

196 − 50 =

K = ME + W

K_f = 146J

$K = \frac{1}{2}mv^2$

V = 12,083 m

$146 = \frac{1}{2}(2)v^2$

$146 = v^2$

Problem 6.) Gravitation

Crab nebula mass = $1.99 \cdot 10^{30}$ kg

Neutron star radius = 12 km

Find gravitational acceleration

12km = $1.2 \cdot 10^4$ m

Formulas

$G = 6.67 \cdot 10^{-11}\ Nm^2/kg^2$

$g = \frac{GM}{r^2} = \frac{(6.67 \cdot 10^{-11})(1.99 \cdot 10^{30}\ kg)}{(1.2 \cdot 10^4 m)^2}$

$\frac{1,327\,33E20}{1.44E8} = 9,217\,569\,444$

$g = 9,217\,569\,444\ E11$

5 − 0.506 = 2.494

Chad Saunders

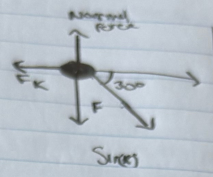
Problem set 1

a.) F= 15N

Mass Cart = 12.5

Kinetic Friction = 104.249

$\mu_K = 0.851$



B.)

$F \cdot m \cdot g =$

$0.851 \cdot 12.5 \cdot 9.8 = 104.24 \text{ kg m/s}^2$

C.) a=?

$(a) \cdot 12.5 = F \cdot \cos(30) - 104.249$

$\frac{12.5a}{12.5} = \frac{104.2 + 75 - 104.5}{12.5}$

$a = -7.3 \text{ m/s}^2$

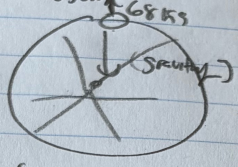
Problem set 2 → Uniform Circular Motion

Data

Mass = 68kg a_c 111.4

Normal force = 555 N

Gravity = $9.8a \downarrow = \frac{m}{s^2}$



a.) What is the value of the force

$555 + 9.8a = 9.8 \cdot 68$

$-555 \quad\quad -555$

$\frac{9.8a}{9.8} = \frac{666.4}{9.8}$

$a = 11.36$

B.)

$m(g+F)$

$(68)(9.8 + 11.36) = 1438.88 \text{ N}$

C.)

?

Confused

Problem set B). Work

Formula = $F_{cy} = k_1^2 - k_2 x^2$

Data

$k_1 = 150 \text{ N}$

$k_2 = 190 \text{ N/m}^2$

$d = 0.2 = m$

$X = 0.2$

$F \downarrow \quad \int_{x=1}^{N+1} \quad N+1$

$\int k_1 x - k_2 x^2 = = \frac{150}{2} x^2 - \frac{190}{3} x^3$

$75(0.2)^2 m - \frac{190}{3} x^3 =$

$\frac{15}{3} - \frac{63.3(0.2)^3}{3}$

$15 - 0.506$

$3 - 0.506 = 2.494$

Work done by the Athlete = 2,494 N